

Gautham G

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SUMMARY

Senior Generative AI Architect driving end-to-end AI solutions for healthcare and enterprise. Senior-level expertise leading cross-functional teams to design and deploy scalable GenAI platforms using TensorFlow, Keras, LangChain and LlamaIndex. Delivered a cloud-native generative-AI platform that cut model deployment time by 30% and enabled real-time analytics for a major government client. Seeking to leverage deep learning and LLMops experience to accelerate AI innovation and business impact.

TECHNICAL SKILLS

- **Programming & Scripting:** Python, NumPy, Pandas, Matplotlib, PySpark, Java, JavaScript, TypeScript, Shell Scripting, Bash, PowerShell, C++, SQL, HTML5, JSON, YAML
- **Web & UI Technologies:** React.js, Angular, JavaScript, HTML5, CSS3, Bootstrap, Node.js, Flask, Django, FastAPI, REST, GraphQL
- **Frameworks & Libraries:** Spring Boot, Spring MVC, Hibernate, TensorFlow, PyTorch, Scikit-learn, Keras, OpenCV, Transformers (Hugging Face), LangChain, CrewAI, LangGraph, AutoGPT, XGBoost
- **Cloud Platforms & Services:** AWS (SageMaker, Lambda, EC2, S3, EKS, Glue, Bedrock, Step Functions, CloudWatch, IAM, Athena, Redshift, DynamoDB), Azure (Azure ML, Azure OpenAI, AKS, Blob Storage, Azure Functions, Azure Monitor, Cognitive Search, AI Studio), GCP (Vertex AI, GKE, Cloud Storage, BigQuery, Dataflow, AI Platform)
- **DevOps & Infrastructure as Code:** Docker, Kubernetes (EKS, AKS, GKE), Terraform, Helm, CloudFormation, CDK, Jenkins, GitHub Actions, GitLab CI, ArgoCD, MLflow, DVC, Vault, Prometheus, Grafana, ELK Stack, Ansible
- **Databases & Storage:** PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, Cassandra, Snowflake, Amazon Aurora, DynamoDB, BigQuery, Azure Cosmos DB, Iceberg, Redis, Hive
- **MLOps & Model Deployment:** MLflow, Kubeflow, SageMaker Pipelines, FastAPI, TorchServe, TensorFlow Serving, ONNX, CI/CD for ML, Model Registry, Feature Store, Model Monitoring, Databricks, Unity Catalog, Colibra, Privacera
- **Machine Learning & AI:** Supervised/Unsupervised Learning, Deep Learning, NLP, Computer Vision, OCR, LLMs (GPT, LLAMA, Claude), Transformers, GANs, VAEs, ViT, YOLOv5/v8, TrOCR, OpenAI API, Bedrock, Anthropic API, Prompt Engineering, RAG, AutoML
- **Data Engineering & Pipelines:** Apache Spark, Airflow, Kinesis, Glue, Kafka, Kafka Streams, Apache Beam, Dataflow, ETL/ELT (dbt, Airbyte, Informatica), ETL (Extract, Transform, Load), Data pipelines, Data architecture, Data models and structures, Machine Learning model deployment, Batch and real-time data pipelines, Metadata management, Data lineage, Data quality monitoring
- **Monitoring & Logging:** CloudWatch, Azure Monitor, Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), Fluentd, AlertManager
- **Operating Systems:** Linux (Ubuntu, Amazon Linux, CentOS, RHEL), Windows, macOS
- **Tools & IDEs:** VS Code, PyCharm, IntelliJ IDEA, Eclipse, Xcode, JupyterLab, Postman, Toad, SQL Developer, Git, GitHub, GitLab, Bitbucket

TECHNICAL EXPERIENCE

Digitech(IBM- state of arizona)

Jul 2025 - Present

Senior AI Engineer – Cloud Platform & AI Systems Engineering

Phoenix, AZ

Leading AI innovation and data unification for the Arizona Department of Child Safety a critical state government platform. Architecting a cloud-native identity consolidation system using Informatica MDM and Customer 360 to produce trusted Golden Records from 15M+ citizen records across 10+ source systems. Concurrently designing and productionizing GenAI solutions including an LLM-powered allegation classification system and agentic workflows, while building scalable Azure data pipelines for CDC, medallion architecture, and real-time analytics.

- Designed and productionized a GPT-4 and LangChain-powered allegation classification system for the state child safety platform, applying risk-based tiering (likelihood of harm to child) across 3M+ active case records and 394K historical records — achieving 95% classification accuracy at production scale.
- Engineered end-to-end GenAI pipelines to ingest unstructured allegation narratives, extract structured risk signals via prompt-chained LLM calls, and output tiered risk scores — enabling caseworkers to prioritize high-harm cases faster and reducing manual triage effort across a high-stakes government child welfare system.
- Built LangChain agentic workflows with multi-step reasoning chains to assess allegation severity, cross-reference historical case patterns, and generate risk justification summaries — integrating short-term conversational memory and long-term vector-backed case history retrieval to maintain context across complex, multi-document case files.
- Implemented RAG pipelines to index 394K historical child safety records into a vector store, enabling the classifier to retrieve semantically similar prior cases at inference time — improving tier prediction consistency and grounding LLM outputs in verified precedent rather than hallucination.
- Optimized prompt templates with metadata chaining, intent scoring, and few-shot examples to improve risk-tier boundary precision; iterated on retrieval strategies and re-ranking logic to push classification accuracy to 95% — directly reducing

misclassified high-risk cases and improving caseworker response prioritization.

- Collaborated with legal, compliance, and child safety policy teams to implement responsible AI guardrails — including hallucination filters, explainability logs, and consent tracing — ensuring the LLM classification system met state data governance and child-safety privacy regulations before production deployment.
- Led end-to-end POC and requirements discovery for an enterprise MDM initiative across 10+ source systems (Dynamics 365, Dataverse, Synapse, legacy SQL) identifying citizen identity attributes (Name, DOB, Race, Ethnicity, Language, Tribal Affiliation, Identifiers) and designing an Informatica MDM + Customer 360 architecture that consolidated 15M+ records into trusted Golden Records, successfully transitioning from POC to full production deployment.
- Designed and executed fuzzy match-and-merge rules (same name + DOB, same name + identifier, similar name + DOB) across Guardian Person, Intake Person, and MEP Person entities deduplicating 15M raw records to 8M consolidated Golden Records, a 47% reduction in data redundancy, with full batch processing of 20M records and daily incremental loads of 100K capturing all MDM state changes and survivorship decisions.
- Architected and delivered enterprise Azure data pipelines using Azure Data Factory, Databricks (PySpark), and Synapse Analytics — processing 100M+ records across 2,000+ tables through a Bronze / Silver / Gold medallion architecture, improving data processing efficiency by ~30% and enabling high-performance analytics and Power BI reporting for child safety leadership.
- Built CDC pipelines using Azure Data Lake, Databricks, and stored procedures with checksum-based change detection — ensuring only mutated records are reprocessed on daily 100K incremental loads, reducing redundant compute while maintaining full dimensional shift tracking and audit history across 8M consolidated person records.
- Designed event-driven CDC architecture using Azure Event Hub capturing MDM business events and routing them via Informatica streaming tasks, reducing synchronization latency by ~40% versus the existing batch model; executed POC validating sub-minute latency for person record updates, informing the real-time MDM streaming roadmap.
- Implemented data governance, metadata management, and lineage tracking frameworks ensuring HIPAA, FedRAMP, and enterprise data policy compliance; optimized Synapse SQL queries improving large-volume query performance by ~35%; deployed enterprise REST API integrations across 3 state agency systems improving data availability by 30%.
- Implemented CI/CD pipelines using Azure DevOps and GitHub Actions with automated validation checks — reducing deployment errors by ~30% and production issues by 40% across dev -> QA -> prod environments; integrated Azure Key Vault and Logic Apps for secure credential management and pipeline orchestration.

Client George Mason University, IBM

Jan 2024 - Jul 2025

Gen AI Engineer – Deep Learning & Internet Infrastructure Security

Fairfax, VA

Contributed to the redevelopment of the global DNSSEC Deployment Maps system for the Center for Assurance Research and Engineering (CARE) and the Measurable Security Lab (MSL) at GMU. Helped transition and modernize a 17-year legacy system (SecSpider), built web and agent-based interfaces for monitoring DNSSEC-enabled domains across the world.

- Modernized a 17-year legacy DNSSEC monitoring system (SecSpider) by redesigning it into a scalable microservice-based architecture using Angular, React, Node.js, and Docker, reducing deployment complexity by 60% and improving system reliability across 1M+ global TLDs.
- Built full-stack web applications (Angular, React, Node.js) to visualize and manage global DNSSEC zone data, enabling real-time monitoring of 100K+ domains and reducing manual analysis time by 40%.
- Containerized services using Docker, standardizing development and production environments, which improved deployment consistency and reduced environment-related failures by 50%.
- Migrated and optimized legacy Perl-based polling scripts to automate DNS zone discovery, TLD uptime monitoring, and DNSSEC signature validation, increasing polling efficiency by 35% and improving validation accuracy across 10k+ TLD registries.
- Engineered MySQL-backed telemetry services to log and analyze historical DNS zone data enriched with security metadata, supporting longitudinal analysis of 20 years of DNSSEC deployment trends.
- Developed interactive data visualizations using D3.js, generating dynamic graphs and bar charts for DNSSEC adoption and TLD health metrics, improving stakeholder reporting clarity and reducing report preparation time by 30%.
- Supported telemetry ingestion pipelines for the Internet Namespace Security Observatory (INSO), enabling continuous ingestion of 1million+ DNS records per month for research and policy analysis
- Designed and deployed an AI-driven DNSSEC analytics chatbot using IBM Watson Orchestrate and IBM watsonx.ai, enabling natural-language querying of DNS telemetry and reducing researcher query turnaround time by 50–70%.
- Implemented Retrieval-Augmented Generation (RAG) pipelines to index structured and semi-structured DNS telemetry data, improving answer accuracy by 50% vs baseline LLM responses.
- Integrated multi-source datasets (INSO databases, DNSSEC Deployment Maps, historical telemetry repositories) into a unified AI-ready knowledge layer supporting cross-dataset security insights.
- Built Python-based ETL pipelines with dbt to transform raw DNS telemetry into structured document corpora for RAG ingestion, automating data-to-insight workflows and reducing manual preprocessing effort by 60%.
- Automated generation of Q&A responses, policy summaries, and security trend reports, including AI-generated graphs and adoption analytics, accelerating report generation cycles from days to minutes.
- Integrated LangChain-based AI agents with Claude MCP and local DNSMap datasets, enabling contextual DNSSEC trend analysis and advanced telemetry exploration via conversational interface.
- Collaborated with CARE leadership, faculty researchers, and cybersecurity advisors to align AI outputs with academic research and

security policy, resulting in the adoption of AI-generated insights across multiple research projects and policy briefs

Virginia Tech

Jul 2023 - Dec 2023

Gen AI Research Intern – Deep Learning & Internet Infrastructure Security

Fairfax, VA

Contributed to research on biometric authentication using hyperspectral imaging and transformer-based architectures. Built and evaluated deep learning pipelines comparing CNN models with modern vision transformers like ViT and Swin, alongside a custom SpectralFormer architecture.

- Developed a hyperspectral palmprint verification model using SpectralFormer, Swin Transformer, and Vision Transformer (ViT), achieving higher verification accuracy than baseline CNN models
- Compared traditional CNN-based classification pipelines with transformer-based architectures across accuracy, F1, SSIM, and convergence time, revealing that transformer models achieved higher accuracy and faster convergence
- Applied PCA and KMeans clustering to reduce dimensionality of hyperspectral data and accelerate model training.
- Authored and tuned a custom contrastive loss function to improve generalization across multimodal biometric pairs.
- Automated model training and evaluation workflows, enabling reproducible benchmarking experiments across biometric datasets (face, palm, fingerprint).
- Defined and executed unit, integration, and system-level test cases for data ingestion and alerting workflows; implemented automated alert triage pipelines to streamline incident response.
- Delivered training sessions and demos on AI-assisted metadata querying tools, empowering users to explore data lineage and system health insights using natural language

Informatica LLC

Jan 2021 - Jul 2023

Senior AI Cloud Engineer – Cloud Platform Engineering (IICS & OpsInsight)

Bengaluru, India

Spearheaded development of highly scalable telemetry, monitoring, and ML-driven recommendation pipelines within Informatica's IICS OpsInsight platform to enable real-time alerting, intelligent diagnostics, and operational intelligence across hybrid cloud environments (AWS, Azure, GCP). Designed microservices and data workflows to collect and analyze operational metadata from 10K+ Secure Agents, generating CLAIRE AI-based insights and automated workflow recommendations. Enabled platform-wide visibility, reduced downtime, and delivered actionable guidance to optimize data engineering, ingestion, and integration workflows at scale.

- Designed and implemented OpsInsight microservices to collect telemetry from Secure Agents, monitor IICS environments, and generate insights using CLAIRE AI.
- Built ML-powered recommendation pipelines using CLAIRE AI to analyze telemetry data and suggest workflow optimizations, alert thresholds, and agent configuration tuning.
- Developed behavioral models and pattern recognition logic to proactively recommend data remediation actions and auto-resolve common ingestion issues.
- Built and optimized complex ANSI SQL queries using Hue (Impala Editor) for telemetry and diagnostic data; leveraged pivots, joins, and CASE logic to support anomaly detection and root cause analysis.
- Developed Kafka-based ingestion pipelines and Airflow DAGs to orchestrate operational metadata collection across all major collectors including: Domain Configuration Collector - fetched details on domains, nodes, and services every 24 hours. Domain Health Resource Usage Collectors - streamed availability, CPU, and memory metrics every 5-60 mins. Data Engineering & Data Quality Collectors - gathered job-level execution stats from Hadoop-based environments. PowerCenter Repository Collector – pulled workflow/session metrics from JDBC-based repositories hourly.
- Built alert orchestration services to configure and dispatch: Infrastructure Alerts for Secure Agent and domain-level failures. Data Integration Alerts for job failure, threshold breaches, and auto-remediation (e.g., job stop/restart). Data Ingestion and Replication Alerts for job state anomalies across ingestion assets. PowerCenter Alerts for workflow-level exceptions, including CLAIRE anomaly-based detections.
- Developed interactive dashboards and heatmaps to visualize trends, usage, and SLAs across services such as Application Integration, Data Quality, Data Profiling, and PowerCenter domains.
- Built and containerized microservices for agent onboarding, collector scheduling, domain registration, and alert subscriptions using Spring Boot and Docker.
- Designed and scaled GenAI solutions across banking and healthcare clients using TensorFlow, Keras, and Elygah, with cloud orchestration across AWS and Azure, enhancing model accuracy for critical use cases
- Implemented generative modeling frameworks (VAE + autoregressive fusion) for credit risk and healthcare analytics, reducing data acquisition costs by 50%.
- Served as enterprise lead on LLMOps pipeline architecture, introducing feedback loops and model caching for large-scale inference APIs.
- Oversaw the creation of reusable prompt libraries and modular decision agents, deployed through microservice containers with Helm and Kubernetes.
- Mentored junior data scientists and engineers on GenAI patterns, model explainability, and evaluation metrics across multi-modal use cases.
- Implemented monitoring and evaluation frameworks for indexing quality, latency, and relevance scoring to ensure reliability in production environments.
- Supported development of an LLM-powered RAG chatbot using Azure AI Search and OpenAI/GPT, automating customer support and reducing response latency
- Explored advancements in vector databases, LLM fine-tuning, prompt engineering, and hybrid retrieval, delivering three prototypes

that informed the product roadmap

- Integrated Prometheus, Grafana, Fluentd, ELK, and AlertManager for real-time observability and service-level monitoring across cloud regions.
- Automated multi-cloud CI/CD workflows for 75+ services using Jenkins, GitLab CI, Helm, Spinnaker, Terraform, and AWS CDK.
- Designed sandboxed views and temporary tables to stage and transform operational metadata for downstream analytics and alerting workflows.
- Built alert orchestration services that dispatched intelligent infrastructure and data-job alerts with AI-based scoring and prioritization, improving alert relevance and reducing noise

Nokia

Jan 2019 - Dec 2020

AI Engineer - Fixed Network Management Solutions (Networks & Solutions Group)

Bengaluru, India

Worked within the Fixed Network Management Solutions team to modernize broadband provisioning and network control services. Contributed to cloud native automation platforms (Altiplano, Access Management System) to manage provisioning, network optimization, and lifecycle operations for FTTH, GPON, DSL, and cable-based home networks. Developed back-end services, CI/CD pipelines, and HA infrastructure on Kubernetes and multicloud, enabling rapid scaling and resilient network management.

- Developed RESTful microservices using Java and Spring Boot that automated fixed network provisioning and decommissioning of home broadband connections, accelerating the provisioning workflow
- Supported legacy SOAP integrations using JAX-WS and built modern Cloud-Native Network Functions (CNFs) for Kubernetes deployments, ensuring a seamless migration to cloud-native architecture
- Used Hibernate with Oracle/PostgreSQL and second-level caching to improve data retrieval for network element inventory and path computation.
- Built UI modules with Angular and jQuery that delivered real-time dashboards for network provisioning status and fault alerts, enhancing operational visibility
- Containerized and deployed services to Kubernetes with autoscaling and continuous delivery pipelines using Jenkins and Docker, enabling rapid scaling of network-management components
- Integrated persistent storage by working with Elasticsearch/OpenSearch and Neo4J to manage data in highly available, scalable stores for AI-based applications, improving data accessibility and reliability
- Coordinated test and deployment activities with QA and configuration-management teams, writing and executing unit tests and ensuring smooth releases, which reduced post-deployment defects
- Monitored and maintained service health using Prometheus and set up automated alerts for system bottlenecks.
- Wrote Python and Bash scripts to automate service restarts, log collection, and remote deployment tasks, cutting manual effort and speeding up incident response
- Performed load and SLA testing using JMeter and Selenium to ensure telecom-grade performance and reliability.

Orbysol Systems

Jul 2018 - Jan 2019

Junior Software Engineer – Product Engineering

Hyderabad, India

Worked on a healthcare analytics project to build mobile and web applications that collect and display real-time health metrics such as heart rate, breathing rate, and SpO2 levels from Bluetooth-connected body sensors. Designed and developed both iOS and web-based dashboards using Swift and JavaScript, and implemented backend services using Java Spring and MySQL. Helped create a user-friendly interface that allowed users to view daily and weekly trends.

- Developed a Bluetooth-enabled iOS app using Swift that collected live biometric data (heart rate, SpO2, respiratory rate) from wearable devices, enabling users to monitor health in real time
- Designed a responsive web dashboard with AJAX, JavaScript, HTML, and CSS to display personalized health metrics, improving user access to their data across devices
- Built RESTful APIs using Spring MVC and Hibernate that connected the front end to a MySQL database, allowing secure storage and retrieval of patient data
- Created back-end logic to compute daily and weekly health summaries for each user, providing clear information about health trends
- Managed MySQL database schemas to support biometric data logging and user-specific metric tracking, ensuring data integrity and scalability
- Utilized JAXB for XML processing and integrated SOAP-based services where legacy compatibility was required, extending system interoperability
- Implemented Git and GitLab CI for source control and automated deployments, reducing release cycle time and minimizing manual errors
- Collaborated with the QA team to test mobile and web features in multiple environments, enhancing overall product quality

Certifications

- Google Professional Machine Learning Engineer Certification
- OCI data Science
- OCI Generative AI

- Crew AI Multi Agent Systems
- Hugging Face Fundamentals of agents

Education

Virginia Tech

Masters, Computer Science

•GPA: 4

Aug 2023 - Dec 2024

Blacksburg, VA